

Rank-four finite T-data

A computer-assisted classification

5 September 2026

Abstract

With identity symmetrizer and diagonal leading matrix, there are thirty-seven indecomposable rank-four families, up to rational rescaling of time, species shifts, index permutations and sign exchange. We give explicit representatives, their matrices at $z = 1$, and exact periodicity certificates. A bound of 15 on off-diagonal coefficient sums reduces necessity to a finite enumeration, followed by unbounded linear-arithmetic feasibility checks. A leading-term obstruction excludes a remaining chain pattern. All surviving lift spaces consist precisely of rescalings and shifts of their representatives.

1 The classification

The scope is the identity-symmetrizer, diagonal-leading-matrix setting of Mizuno's rank-two paper [1]. Throughout, the rank is the number of species in the difference equation; the reconstructed quiver can have more vertices.

Definition 1.1. Let $r_1, \dots, r_4$ be positive integers and let

$$N_0(z) = \text{diag}(1 + z^{r_1}, \dots, 1 + z^{r_4}), \quad A_{\pm}(z) = N_0(z) - N_{\pm}(z).$$

The entries of $N_{\pm}$ have nonnegative integer coefficients. In entry (i, j) , every exponent p satisfies $0 < p < r_i$, and the two signs have disjoint support. Require the symplectic identity

$$A_+(z)A_-(z^{-1})^T = A_-(z)A_+(z^{-1})^T. \quad (1)$$

The datum is indecomposable if no simultaneous index permutation makes both matrices block diagonal with two nonempty blocks. Finite type means periodicity of the associated universal Y-system, as in [1, Definition 1.4].

Theorem 1.2 (Computer-assisted classification). *An indecomposable datum in Definition 1.1 is of finite type if and only if, after an index permutation and possibly exchanging the signs,*

$$A_{\pm}(z) = E(z)B_{\pm}^{(k)}(z^{\lambda})E(z)^{-1}, \quad E(z) = \text{diag}(z^{s_1}, z^{s_2}, z^{s_3}, 1), \quad (2)$$

for exactly one of the thirty-seven representatives in Section 7. Here $\lambda \in \mathbb{Q}_{>0}$ and $s_1, s_2, s_3 \in \mathbb{Q}$. The parameters are precisely those for which all transformed exponents are integral and the support inequalities hold.

Thus a representative term z^p in entry (i, j) becomes $z^{\lambda p + s_i - s_j}$, and $r_i = \lambda r_i^{(k)}$, with $s_4 = 0$. This specifies all admissible polynomial lifts at arbitrarily large delays. The thirty-seven representatives also belong to distinct classes under change of slices in [1, Section 3.1]; see Section 5.

Corollary 1.3. *For the rank-four data of Definition 1.1, finite type is equivalent to the existence of a row vector $v > 0$ with $vA_+(1) > 0$ and $vA_-(1) > 0$.*

For decomposable data, apply the theorem blockwise. The remaining rank partitions are $3 + 1$, $2 + 2$, $2 + 1 + 1$ and $1 + 1 + 1 + 1$, using the rank-three, rank-two and rank-one classifications. Each component may have its own time scale; a common period is a common multiple of the component periods. The statements concern $D = I_4$ and diagonal N_0 , rather than the full definition allowing other symmetrizers and leading permutations.

2 A finite reduction without a delay cutoff

Write $P = N_+(1)$ and $M = N_-(1)$, so that $A_{\pm}(1) = 2I_4 - P, 2I_4 - M$. The necessary simultaneous positivity theorem is [1, Lemma 3.3], citing the arbitrary-rank theorem [2, Theorem 5.5]. Hence a finite-type datum has a row vector $v > 0$ with

$$v(2I_4 - P) > 0, \quad v(2I_4 - M) > 0. \quad (3)$$

In particular $P_{ii}, M_{ii} \in \{0, 1\}$, and both $2I_4 - P$ and $2I_4 - M$ have positive principal minors. Indeed, scaling by v makes their transposes strictly diagonally dominant Z-matrices. Every principal submatrix retains this property and has positive determinant.

Lemma 2.1. *For an indecomposable datum, the graph with arrows $i \rightarrow j$ when $i \neq j$ and $P_{ij} + M_{ij} > 0$ is strongly connected.*

Proof. If the graph is reducible, order its strongly connected components so that A_+ and A_- are block lower triangular. The matrices have constant term I_4 . Thus $K = A_+^{-1}A_-$ exists over $\mathbb{Q}(z)$ and is regular at zero. Equation (1) gives $K(z) = K(z^{-1})^T$, so its block lower triangular form is block diagonal.

Suppose that some entry between distinct blocks of N_+ or N_- is nonzero, and let $h > 0$ be the smallest degree of such an entry. In $K = I_4 + A_+^{-1}(N_+ - N_-)$, the coefficient of degree h between blocks is the corresponding coefficient of $N_+ - N_-$. Every other contribution has a crossing factor of degree at least h and a further factor of positive degree. Disjoint support makes one degree- h crossing coefficient nonzero, contradicting block diagonality of K . $\square$

Lemma 2.2 (Coefficient bound). *Every finite-type indecomposable datum satisfies $P_{ij}, M_{ij} \leq 15$ for $i \neq j$.*

Proof. Let $w_{ij} = \max(P_{ij}, M_{ij})$. Equation (3) gives $w_{ij}v_i < 2v_j$. Every arrow belongs to a simple directed cycle of length $\ell \leq 4$, by Lemma 2.1. Multiplication along the cycle gives $\prod w_{ij} < 2^\ell$. All its weights are positive integers, so each is at most $2^\ell - 1 \leq 15$. $\square$

Lemma 2.3 (Parity). *Each diagonal entry of $(2I_4 - P)(2I_4 - M)^T$ is even.*

Proof. The diagonal Laurent polynomial $(A_+A_-^*)_{ii}$ is reciprocal, where $F^*(z) = F(z^{-1})^T$. Its constant coefficient is 2: the leading diagonal factors contribute 2, the mixed leading/support terms contribute zero, and disjoint support gives $[z^0] \sum_j (N_+)_{ij}(z)(N_-)_{ij}(z^{-1}) = 0$. Every nonconstant coefficient of a reciprocal Laurent polynomial occurs in an equal pair. Evaluation at 1 is therefore even. $\square$

How the constants are enumerated

Relabel a vector in (3) so that $v_1 \geq v_2 \geq v_3 \geq v_4$. For either $C = P$ or $C = M$, its j th column then satisfies

$$\sum_{i < j} C_{ij} \leq 1 - C_{jj}.$$

Thus, above the diagonal, column j is either zero or has a single 1; the latter requires $C_{jj} = 0$. There are $\prod_{j=1}^4 (j+1) = 120$ choices of the diagonal and strict upper triangle of each matrix. Exchanging the signs leaves $120 \cdot 121/2 = 7,260$ unordered pairs of these choices to process.

The program completes the rows in order. When row j is reached, it enumerates the unknown entries of P in $\{0, \dots, 15\}$ and uses

$$(2I_4 - P)(2I_4 - M)^T = (2I_4 - M)(2I_4 - P)^T$$

to solve uniquely for the unknown entries of that row of M . The coefficient matrix is the already completed leading principal block of $2I_4 - P$, whose determinant is positive. Adjugates, divisibility checks, cycle bounds and principal minors are computed with integers. No solution can be lost by solving this nonsingular linear system.

Completed pairs are tested for strong connectivity and ordered simultaneous positivity, and then identified under all 24 index permutations and sign exchange. For positivity, the C++ enumerator constructs extreme rays of

$$\{v : v \geq 0, v(2I_4 - P) \geq 0, v(2I_4 - M) \geq 0, v_i \geq v_{i+1}\}.$$

Each extreme ray is obtained from three independent active constraint normals. The cone is pointed; the sum of its rays is strict in the required inequalities exactly when a suitable v exists. A separate rational linear solver supplies and verifies an integer positive vector for every retained pair, without imposing the ordering on its canonical representative.

Exact computation 2.4. The enumeration gives 4,865 distinct constant pairs. The independent verifier checks their symplectic identities, all principal minors, strong connectivity, canonical forms and explicit simultaneous positive vectors. Lemma 2.3 excludes 4,405 pairs, leaving 460.

As a control on completeness of the new enumeration method, its rank-three version reproduces exactly the entire set of 180 constant pairs in the rank-three calculation, not merely its cardinality.

3 The chain obstruction

The following elementary exclusion is used before the remaining feasibility checks. It also gives a direct proof for the largest surviving coefficient pattern in the constant enumeration.

Lemma 3.1. *No datum of Definition 1.1 has P, M of the form*

$$P = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ u & v & w & 0 \end{pmatrix}, \quad M = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & t & 0 \end{pmatrix}, \quad u, v, w, t \in \mathbb{Z}_{\geq 0}.$$

Proof. The (1,2) and (2,3) symplectic equations first imply $r_1 = r_2 = r_3 = R$ and

$$(N_-)_{12} = z^a, \quad (N_-)_{21} = z^{R-a}, \quad (N_-)_{23} = z^c, \quad (N_-)_{32} = z^{R-c}.$$

For example, the first equation is equality of two translated binomials; their lowest and highest exponents give the asserted equalities. Write $(N_-)_{14} = z^b$, and define the Laurent polynomials

$$\begin{aligned} q &= z^R, & F &= 1 + q, & H &= z^b(1 + z^{-r_4}), \\ X &= (N_+)_{41}(z^{-1}), & Y &= z^a(N_+)_{42}(z^{-1}), \\ Z &= z^{a+c}(N_+)_{43}(z^{-1}), & W &= z^{a+c}(N_-)_{43}(z^{-1}). \end{aligned}$$

The (1,4), (2,4) and (3,4) equations give, respectively,

$$Y = FX + H, \quad FY = qX + Z, \quad F(Z - W) = qY. \quad (4)$$

Consequently $Z = (1 + q + q^2)X + FH$. The polynomials X, H have nonnegative coefficients and $H \neq 0$. The first two expressions therefore show that Y and Z have the same lowest exponent d . Since $R > 0$, the lowest exponent of qY is $d + R > d$. The last equation in (4) forces W to have lowest exponent d , with the same nonzero coefficient there as Z . This gives a common exponent in $(N_+)_{43}$ and $(N_-)_{43}$, contradicting disjoint support. $\square$

The lemma applies to constant IDs 22, 35, 44. ID 35 has already been removed by parity. Thus 458 pairs remain for the unbounded polynomial calculation.

4 All polynomial lifts and their periods

For each constant pair, introduce four delay variables and one exponent variable for every unit of coefficient mass. Equal exponents of the same sign are allowed, so coefficients greater than one are included. Exponents within an entry are ordered, opposite signs have distinct exponents, and every exponent is strictly between zero and its row delay.

Expand $A_+A_+^* - A_-A_+^*$. In each entry, equality to zero is equality of two finite multisets of integral linear forms in these variables. The main encoding compares multiplicities at every positive-side exponent; equality of the total sizes follows from the constant-matrix identity. A second encoding sorts the two lists through exact comparison networks and equates them. Both describe the full polynomial identity, including coincident monomials. There is no upper bound on delays or exponents.

Most queries use integer variables. A homogeneous query can also be solved over the reals with $\sum r_i = 1$. Each Boolean branch is a rational linear system with strict inequalities. A nonempty such system has a rational point; clearing denominators yields an integer solution of the original homogeneous query. Conversely any integer solution can be normalized. This justifies the rational arithmetic used for the exclusion of ID 1117.

Exact computation 4.1. Of the 458 remaining constant pairs, 421 have no polynomial lift and 37 have a lift. For each survivor, all admissible lifts lie in a single four-dimensional rational linear subspace. Its generators are the representative's exponent vector and the three relative species shifts in (2). There are no unresolved solver answers.

To obtain a subspace, match equal exponents on the two sides of each Laurent identity at a satisfying point. The matching imposes homogeneous linear equations that imply the entire identity. The verifier reduces the exponent forms modulo the resulting row-reduced matrix and checks equality of the two multisets symbolically. It also checks the dimension and the four generators, then asks whether any admissible lift lies outside that subspace. Every such coverage query is UNSAT. This proves necessity in Theorem 1.2 for every delay.

Sufficiency

For each representative, reconstruct the skew-symmetric exchange matrix on $\{(i, p) : 1 \leq i \leq 4, 0 \leq p < r_i\}$ using [1, equation (2.2)]. A time step mutates the four vertices $(i, 0)$ and relabels $(i, p) \mapsto (i, p - 1 \bmod r_i)$. The verifier checks their commutativity, the return of the exchange matrix and inverse-step consistency. Starting from the identity principal C-matrix, integer mutation gives a negative permutation matrix in both directions for all 37 representatives. Proposition 2.7 of [1] then proves finite type in every semifield. The lengths h_+, h_- and the first positive return Ω of the labelled C-matrix are listed in Section 6. These are exact seed certificates, rather than numerical tests of a few initial values.

Integral species shifts reindex solutions, and substituting z^m splits time into m independent residue classes. Both operations preserve finite type in both directions. For rational parameters in (2), clear their denominators by an integer L . The transformed datum at z^L is an integral species shift of the representative at $z^{L\lambda}$. The integral argument therefore proves sufficiency throughout each family. Combined with (3) and Computation 4.1, this also proves Corollary 1.3.

5 Equivalences and the degree-two check

Specialization at $z = 1$ is invariant under rescaling and species shifts. The 37 canonical constant pairs are distinct under index permutations and sign exchange. Thus the explicit families in (2) do not overlap.

For change of slices, integral dilations repeat a slice sequence, while species shifts move mutation events and the initial cut. The admissible shift domain is convex. Along a straight path of shifts, the event order changes only at coincident times; after clearing denominators these are simultaneous mutations in a valid datum and they commute. Moving the initial cut rotates the sequence. Rational rescalings and admissible shifts thus give changes of slices.

Distinctness under this larger description is checked directly on connected quiver slices. For each representative, follow one connected component until the time-step permutation returns it. Its mutation word has four events. The canonical encoding records the exchange matrix and the cycles of the terminal permutation rooted at those events. The program exhausts cyclic rotations and adjacent commuting mutations, including the negated exchange matrix for sign exchange. All 37 canonical minima are distinct. Full encodings are stored, together with hashes; distinctness is checked on the encodings themselves.

Exact computation 5.1 (All delays equal to two). The degree-two subcase has nine classes. Enumeration of all $3^{10} = 59,049$ symmetric signed matrices gives 190 connected labelled pairs satisfying commutation and positive principal minors, and nine orbits. Each has an exact two-sided reddening certificate. They occur in the full catalogue as classes 1, 2, 3, 28, 29, 30, 31, 34, 35.

For this subcase $N_{\pm}(z) = zC_{\pm}$. Comparing Laurent coefficients forces $C_{+} - C_{-}$ to be symmetric; disjoint support then makes each $C_{\pm}$ symmetric. The remaining identity is $C_{+}C_{-} = C_{-}C_{+}$. Positivity bounds every entry by one, so the signed symmetric matrix has ten independent entries in $\{-1, 0, 1\}$. This proves that the degree-two check is exhaustive.

6 Periods and slice sizes

The class number is used throughout the catalogue; ID is the index in the exhaustive constant enumeration. The period Ω is measured in the displayed time coordinate. A slice is one connected component of the reconstructed exchange matrix.

Class	ID	Delays	h_+	h_-	Ω	Slice size
1	1	(2,2,2,2)	2	6	8	4
2	2	(2,2,2,2)	2	5	14	4
3	3	(2,2,2,2)	2	9	11	8
4	4	(3,3,3,2)	3	11	14	11
5	5	(6,6,2,6)	6	20	26	10
6	36	(2,2,2,14)	14	30	44	10
7	46	(5,5,2,2)	5	12	17	14
8	47	(8,8,2,6)	8	22	30	12
9	48	(5,5,2,3)	5	13	18	15
10	56	(10,2,2,8)	10	18	28	11
11	57	(7,2,2,5)	7	12	19	16
12	58	(10,2,2,8)	10	22	32	11
13	65	(2,2,2,4)	6	4	20	5
14	79	(2,2,2,10)	12	10	22	8
15	82	(2,2,2,7)	9	7	16	13
16	122	(3,3,2,2)	3	15	18	10
17	136	(2,2,8,8)	12	30	42	10
18	207	(14,2,6,8)	14	22	36	15
19	413	(2,2,16,12)	42	24	66	16
20	432	(8,8,2,2)	8	18	26	10
21	436	(14,6,2,6)	14	20	34	14
22	437	(8,3,2,3)	8	11	19	16
23	453	(2,12,2,12)	26	18	44	14
24	456	(7,2,3,2)	7	15	22	14
25	469	(8,3,2,5)	8	13	21	18
26	949	(2,12,10,12)	34	18	52	18
27	1262	(3,2,2,2)	3	9	12	9
28	1493	(2,2,2,2)	6	3	18	8
29	1494	(2,2,2,2)	6	3	9	8
30	1834	(2,2,2,2)	3	3	12	4
31	1836	(2,2,2,2)	3	5	16	8
32	1844	(2,2,2,6)	8	8	16	6
33	1848	(2,2,2,10)	18	18	36	8
34	1852	(2,2,2,2)	5	5	20	8
35	1853	(2,2,2,2)	5	3	16	8
36	1902	(2,2,2,3)	5	5	10	9
37	2614	(2,2,6,6)	12	12	24	8

7 The thirty-seven representatives

Each class displays $N_+^{(k)}(z), N_-^{(k)}(z)$ and the full constant matrices

$$A_{\pm}(1) = B_{\pm}^{(k)}(1) = 2I_4 - N_{\pm}^{(k)}(1).$$

These constants are unchanged throughout the corresponding family; relabelling simultaneously permutes rows and columns, and sign exchange interchanges the pair. All entries below are generated from the checked integer witnesses.

Classes 1–2

$$B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, \dots, 1 + z^{r_4}) - N_{\pm}^{(k)}(z).$$

Class 1 (constant ID 1)

$\mathbf{r} = (2, 2, 2, 2)$

$$N_+^{(1)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$N_-^{(1)}(z) = \begin{pmatrix} 0 & 0 & 0 & z \\ 0 & 0 & 0 & z \\ 0 & 0 & 0 & z \\ z & z & z & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 2 & -1 \\ -1 & -1 & -1 & 2 \end{pmatrix}$$

$h_+ = 2, \quad h_- = 6, \quad \Omega = 8.$ Vertices per slice: 4.

Class 2 (constant ID 2)

$\mathbf{r} = (2, 2, 2, 2)$

$$N_+^{(2)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$N_-^{(2)}(z) = \begin{pmatrix} 0 & 0 & 0 & z \\ 0 & 0 & z & 0 \\ 0 & z & 0 & z \\ z & 0 & z & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

$h_+ = 2, \quad h_- = 5, \quad \Omega = 14.$ Vertices per slice: 4.

Classes 3–5

$$B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, \dots, 1 + z^{r_4}) - N_{\pm}^{(k)}(z).$$

Class 3 (constant ID 3)**r** = (2, 2, 2, 2)

$$N_+^{(3)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$N_-^{(3)}(z) = \begin{pmatrix} 0 & 0 & 0 & z \\ 0 & 0 & z & z \\ 0 & z & z & 0 \\ z & z & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & -1 & -1 \\ 0 & -1 & 1 & 0 \\ -1 & -1 & 0 & 2 \end{pmatrix}$$

 $h_+ = 2, \quad h_- = 9, \quad \Omega = 11. \quad \text{Vertices per slice: 8.}$
Class 4 (constant ID 4)**r** = (3, 3, 3, 2)

$$N_+^{(4)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z \end{pmatrix}$$

$$N_-^{(4)}(z) = \begin{pmatrix} 0 & 0 & z & 0 \\ 0 & 0 & z^2 & z^2 + z \\ z^2 & z & 0 & 0 \\ 0 & z & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -1 & 0 \\ 0 & 2 & -1 & -2 \\ -1 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}$$

 $h_+ = 3, \quad h_- = 11, \quad \Omega = 14. \quad \text{Vertices per slice: 11.}$
Class 5 (constant ID 5)**r** = (6, 6, 2, 6)

$$N_+^{(5)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & z^3 & 0 \end{pmatrix}$$

$$N_-^{(5)}(z) = \begin{pmatrix} 0 & z & 0 & 0 \\ z^5 & 0 & 0 & z \\ 0 & 0 & 0 & z \\ 0 & z^5 & z^5 + z & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & 0 & -1 \\ 0 & 0 & 2 & -1 \\ 0 & -1 & -2 & 2 \end{pmatrix}$$

 $h_+ = 6, \quad h_- = 20, \quad \Omega = 26. \quad \text{Vertices per slice: 10.}$

Classes 6–8

$$B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, \dots, 1 + z^{r_4}) - N_{\pm}^{(k)}(z).$$

Class 6 (constant ID 36)**r** = (2, 2, 2, 14)

$$N_+^{(6)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ z^9 + z^5 & z^{11} + z^3 & z^{10} + z^4 & 0 \end{pmatrix}$$

$$N_-^{(6)}(z) = \begin{pmatrix} 0 & 0 & z & 0 \\ 0 & 0 & z & z \\ z & z & 0 & 0 \\ z^7 & z^{13} + z & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ -2 & -2 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -1 & 0 \\ 0 & 2 & -1 & -1 \\ -1 & -1 & 2 & 0 \\ -1 & -2 & 0 & 2 \end{pmatrix}$$

 $h_+ = 14, \quad h_- = 30, \quad \Omega = 44. \quad \text{Vertices per slice: 10.}$
Class 7 (constant ID 46)**r** = (5, 5, 2, 2)

$$N_+^{(7)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z \\ 0 & 0 & z & z \end{pmatrix}$$

$$N_-^{(7)}(z) = \begin{pmatrix} 0 & z & 0 & 0 \\ z^4 & 0 & z^4 + z & z^3 + z^2 \\ 0 & z & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -2 & -2 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

 $h_+ = 5, \quad h_- = 12, \quad \Omega = 17. \quad \text{Vertices per slice: 14.}$
Class 8 (constant ID 47)**r** = (8, 8, 2, 6)

$$N_+^{(8)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z \\ 0 & 0 & z^5 + z & 0 \end{pmatrix}$$

$$N_-^{(8)}(z) = \begin{pmatrix} 0 & z^7 & 0 & 0 \\ z & 0 & z^2 & z^3 + z \\ 0 & 0 & 0 & 0 \\ 0 & z^5 & z^3 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & -2 \\ 0 & 0 & 2 & 0 \\ 0 & -1 & -1 & 2 \end{pmatrix}$$

 $h_+ = 8, \quad h_- = 22, \quad \Omega = 30. \quad \text{Vertices per slice: 12.}$

Classes 9–11

$$B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, \dots, 1 + z^{r_4}) - N_{\pm}^{(k)}(z).$$

Class 9 (constant ID 48)**r** = (5, 5, 2, 3)

$$N_+^{(9)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z \\ 0 & 0 & z^2 + z & 0 \end{pmatrix}$$

$$N_-^{(9)}(z) = \begin{pmatrix} 0 & z^4 & 0 & 0 \\ z & 0 & z^2 & z^3 + z \\ 0 & 0 & z & 0 \\ 0 & z^2 & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}$$

 $h_+ = 5, \quad h_- = 13, \quad \Omega = 18.$ Vertices per slice: 15.**Class 10** (constant ID 56)**r** = (10, 2, 2, 8)

$$N_+^{(10)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z \\ 0 & z^6 + z^2 & z^7 + z & 0 \end{pmatrix}$$

$$N_-^{(10)}(z) = \begin{pmatrix} 0 & 0 & z^2 & z^3 + z \\ 0 & 0 & z & 0 \\ 0 & z & 0 & 0 \\ z^7 & z^4 & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & -2 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -1 & -2 \\ 0 & 2 & -1 & 0 \\ 0 & -1 & 2 & 0 \\ -1 & -1 & 0 & 2 \end{pmatrix}$$

 $h_+ = 10, \quad h_- = 18, \quad \Omega = 28.$ Vertices per slice: 11.**Class 11** (constant ID 57)**r** = (7, 2, 2, 5)

$$N_+^{(11)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z \\ 0 & z^3 + z^2 & z^4 + z & 0 \end{pmatrix}$$

$$N_-^{(11)}(z) = \begin{pmatrix} 0 & 0 & z^2 & z^3 + z \\ 0 & z & z & 0 \\ 0 & z & 0 & 0 \\ z^4 & 0 & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & -2 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -1 & -2 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 2 & 0 \\ -1 & 0 & 0 & 2 \end{pmatrix}$$

 $h_+ = 7, \quad h_- = 12, \quad \Omega = 19.$ Vertices per slice: 16.

Classes 12–14

$$B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, \dots, 1 + z^{r_4}) - N_{\pm}^{(k)}(z).$$

Class 12 (constant ID 58) $\mathbf{r} = (10, 2, 2, 8)$

$$N_+^{(12)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z \\ 0 & z^6 + z^2 & z^7 + z & 0 \end{pmatrix}$$

$$N_-^{(12)}(z) = \begin{pmatrix} 0 & 0 & z^9 + z & z^8 + z^2 \\ 0 & 0 & z & 0 \\ z & z & 0 & 0 \\ 0 & z^4 & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & -2 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -2 & -2 \\ 0 & 2 & -1 & 0 \\ -1 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}$$

 $h_+ = 10, \quad h_- = 22, \quad \Omega = 32. \quad \text{Vertices per slice: 11.}$
Class 13 (constant ID 65) $\mathbf{r} = (2, 2, 2, 4)$

$$N_+^{(13)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z \\ z^2 & z^2 & z^3 + z & 0 \end{pmatrix}$$

$$N_-^{(13)}(z) = \begin{pmatrix} 0 & 0 & z & 0 \\ 0 & 0 & z & 0 \\ z & z & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ -1 & -1 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -1 & 0 \\ 0 & 2 & -1 & 0 \\ -1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

 $h_+ = 6, \quad h_- = 4, \quad \Omega = 20. \quad \text{Vertices per slice: 5.}$
Class 14 (constant ID 79) $\mathbf{r} = (2, 2, 2, 10)$

$$N_+^{(14)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z \\ z^7 + z^3 & z^8 + z^2 & z^9 + z & 0 \end{pmatrix}$$

$$N_-^{(14)}(z) = \begin{pmatrix} 0 & z & 0 & 0 \\ z & 0 & z & 0 \\ 0 & z & 0 & 0 \\ z^5 & 0 & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ -2 & -2 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & 0 \\ -1 & 0 & 0 & 2 \end{pmatrix}$$

 $h_+ = 12, \quad h_- = 10, \quad \Omega = 22. \quad \text{Vertices per slice: 8.}$

Classes 15–17

$$B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, \dots, 1 + z^{r_4}) - N_{\pm}^{(k)}(z).$$

Class 15 (constant ID 82)**r** = (2, 2, 2, 7)

$$N_+^{(15)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z \\ z^5 + z^2 & z^4 + z^3 & z^6 + z & 0 \end{pmatrix}$$

$$N_-^{(15)}(z) = \begin{pmatrix} 0 & z & z & 0 \\ z & z & 0 & 0 \\ z & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ -2 & -2 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 1 & 0 & 0 \\ -1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

 $h_+ = 9, \quad h_- = 7, \quad \Omega = 16.$ Vertices per slice: 13.**Class 16** (constant ID 122)**r** = (3, 3, 2, 2)

$$N_+^{(16)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & z & 0 \\ 0 & 0 & 0 & z \end{pmatrix}$$

$$N_-^{(16)}(z) = \begin{pmatrix} 0 & z & 0 & 0 \\ z^2 & 0 & 0 & z^2 + z \\ 0 & 0 & 0 & z \\ 0 & z & z & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & 0 & -2 \\ 0 & 0 & 2 & -1 \\ 0 & -1 & -1 & 2 \end{pmatrix}$$

 $h_+ = 3, \quad h_- = 15, \quad \Omega = 18.$ Vertices per slice: 10.**Class 17** (constant ID 136)**r** = (2, 2, 8, 8)

$$N_+^{(17)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & z^4 & 0 \\ z^4 & z^5 + z^3 & 0 & z^4 \end{pmatrix}$$

$$N_-^{(17)}(z) = \begin{pmatrix} 0 & z & 0 & 0 \\ z & 0 & 0 & z \\ 0 & 0 & 0 & z \\ 0 & z^7 + z & z^7 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & -2 & 0 & 1 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & 0 & -1 \\ 0 & 0 & 2 & -1 \\ 0 & -2 & -1 & 2 \end{pmatrix}$$

 $h_+ = 12, \quad h_- = 30, \quad \Omega = 42.$ Vertices per slice: 10.

Classes 18–20

$$B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, \dots, 1 + z^{r_4}) - N_{\pm}^{(k)}(z).$$

Class 18 (constant ID 207) $\mathbf{r} = (14, 2, 6, 8)$

$$N_+^{(18)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & z^3 & 0 & z^5 \\ 0 & z^2 & z^3 + z & 0 \end{pmatrix}$$

$$N_-^{(18)}(z) = \begin{pmatrix} 0 & 0 & z^2 & z^7 + z \\ 0 & 0 & z & 0 \\ 0 & z^5 + z & 0 & 0 \\ z^7 & 0 & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & -1 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -1 & -2 \\ 0 & 2 & -1 & 0 \\ 0 & -2 & 2 & 0 \\ -1 & 0 & 0 & 2 \end{pmatrix}$$

 $h_+ = 14, \quad h_- = 22, \quad \Omega = 36.$ Vertices per slice: 15.**Class 19** (constant ID 413) $\mathbf{r} = (2, 2, 16, 12)$

$$N_+^{(19)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ z^3 & z^4 + z^2 & 0 & z^5 + z^3 + z \\ z^8 + z^4 & z^9 + z^3 & z^{11} & 0 \end{pmatrix}$$

$$N_-^{(19)}(z) = \begin{pmatrix} 0 & z & 0 & 0 \\ z & 0 & 0 & z \\ 0 & 0 & z^8 & 0 \\ z^6 & z^{11} + z & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ -1 & -2 & 2 & -3 \\ -2 & -2 & -1 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ -1 & -2 & 0 & 2 \end{pmatrix}$$

 $h_+ = 42, \quad h_- = 24, \quad \Omega = 66.$ Vertices per slice: 16.**Class 20** (constant ID 432) $\mathbf{r} = (8, 8, 2, 2)$

$$N_+^{(20)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z^4 \\ 0 & 0 & 0 & z \\ 0 & 0 & z & 0 \end{pmatrix}$$

$$N_-^{(20)}(z) = \begin{pmatrix} 0 & z & 0 & 0 \\ z^7 & 0 & z^7 + z & z^6 + z^2 \\ 0 & z & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -2 & -2 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

 $h_+ = 8, \quad h_- = 18, \quad \Omega = 26.$ Vertices per slice: 10.

Classes 21–23

$$B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, \dots, 1 + z^{r_4}) - N_{\pm}^{(k)}(z).$$

Class 21 (constant ID 436)

$\mathbf{r} = (14, 6, 2, 6)$

$$N_{+}^{(21)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z \\ 0 & 0 & 0 & z \\ 0 & z^5 & z^5 + z & 0 \end{pmatrix}$$

$$N_{-}^{(21)}(z) = \begin{pmatrix} 0 & z^9 + z & z^3 & z^4 + z^2 \\ z^5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & z^3 & 0 \end{pmatrix}$$

$$A_{+}(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 2 & -1 \\ 0 & -1 & -2 & 2 \end{pmatrix}$$

$$A_{-}(1) = \begin{pmatrix} 2 & -2 & -1 & -2 \\ -1 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

$$h_{+} = 14, \quad h_{-} = 20, \quad \Omega = 34. \quad \text{Vertices per slice: 14.}$$

Class 22 (constant ID 437)

$\mathbf{r} = (8, 3, 2, 3)$

$$N_{+}^{(22)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z \\ 0 & 0 & 0 & z \\ 0 & z^2 & z^2 + z & 0 \end{pmatrix}$$

$$N_{-}^{(22)}(z) = \begin{pmatrix} 0 & z^6 + z & z^3 & z^4 + z^2 \\ z^2 & 0 & 0 & 0 \\ 0 & 0 & z & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$A_{+}(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 2 & -1 \\ 0 & -1 & -2 & 2 \end{pmatrix}$$

$$A_{-}(1) = \begin{pmatrix} 2 & -2 & -1 & -2 \\ -1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

$$h_{+} = 8, \quad h_{-} = 11, \quad \Omega = 19. \quad \text{Vertices per slice: 16.}$$

Class 23 (constant ID 453)

$\mathbf{r} = (2, 12, 2, 12)$

$$N_{+}^{(23)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z^{11} \\ 0 & 0 & 0 & z \\ z^{10} + z^6 + z^2 & z & z^{11} + z & 0 \end{pmatrix}$$

$$N_{-}^{(23)}(z) = \begin{pmatrix} 0 & 0 & z & 0 \\ 0 & z^6 & 0 & 0 \\ z & 0 & 0 & 0 \\ z^8 + z^4 & 0 & 0 & z^6 \end{pmatrix}$$

$$A_{+}(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 2 & -1 \\ -3 & -1 & -2 & 2 \end{pmatrix}$$

$$A_{-}(1) = \begin{pmatrix} 2 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 2 & 0 \\ -2 & 0 & 0 & 1 \end{pmatrix}$$

$$h_{+} = 26, \quad h_{-} = 18, \quad \Omega = 44. \quad \text{Vertices per slice: 14.}$$

Classes 24–26

$$B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, \dots, 1 + z^{r_4}) - N_{\pm}^{(k)}(z).$$

Class 24 (constant ID 456)**r** = (7, 2, 3, 2)

$$N_+^{(24)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z \\ 0 & 0 & 0 & z^2 + z \\ 0 & z & z & 0 \end{pmatrix}$$

$$N_-^{(24)}(z) = \begin{pmatrix} 0 & z^3 & z^5 + z^3 + z & z^4 + z^2 \\ 0 & z & 0 & 0 \\ z^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & z \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 2 & -2 \\ 0 & -1 & -1 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & -3 & -2 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

 $h_+ = 7, \quad h_- = 15, \quad \Omega = 22.$ Vertices per slice: 14.**Class 25** (constant ID 469)**r** = (8, 3, 2, 5)

$$N_+^{(25)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z^2 \\ 0 & 0 & z & 0 \\ 0 & z^3 + z & z^2 & 0 \end{pmatrix}$$

$$N_-^{(25)}(z) = \begin{pmatrix} 0 & z^2 & 0 & z^4 + z \\ 0 & 0 & z^2 + z & 0 \\ 0 & z & 0 & 0 \\ z^4 & 0 & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -2 & -1 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 & -2 \\ 0 & 2 & -2 & 0 \\ 0 & -1 & 2 & 0 \\ -1 & 0 & 0 & 2 \end{pmatrix}$$

 $h_+ = 8, \quad h_- = 13, \quad \Omega = 21.$ Vertices per slice: 18.**Class 26** (constant ID 949)**r** = (2, 12, 10, 12)

$$N_+^{(26)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z^{11} \\ z^7 + z^3 & 0 & 0 & z^9 \\ z^2 & z & z^3 + z & 0 \end{pmatrix}$$

$$N_-^{(26)}(z) = \begin{pmatrix} 0 & 0 & z & 0 \\ 0 & z^6 & 0 & 0 \\ z^9 + z^5 + z & 0 & 0 & 0 \\ 0 & 0 & 0 & z^6 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & -1 \\ -2 & 0 & 2 & -1 \\ -1 & -1 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -3 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

 $h_+ = 34, \quad h_- = 18, \quad \Omega = 52.$ Vertices per slice: 18.

Classes 27–29

$$B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, \dots, 1 + z^{r_4}) - N_{\pm}^{(k)}(z).$$

Class 27 (constant ID 1262) $\mathbf{r} = (3, 2, 2, 2)$

$$N_{+}^{(27)}(z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & z & 0 & 0 \\ 0 & 0 & z & 0 \\ 0 & 0 & 0 & z \end{pmatrix}$$

$$N_{-}^{(27)}(z) = \begin{pmatrix} 0 & 0 & 0 & z^2 + z \\ 0 & 0 & z & 0 \\ 0 & z & 0 & z \\ z & 0 & z & 0 \end{pmatrix}$$

$$A_{+}(1) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$A_{-}(1) = \begin{pmatrix} 2 & 0 & 0 & -2 \\ 0 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

 $h_{+} = 3, \quad h_{-} = 9, \quad \Omega = 12.$ Vertices per slice: 9.**Class 28** (constant ID 1493) $\mathbf{r} = (2, 2, 2, 2)$

$$N_{+}^{(28)}(z) = \begin{pmatrix} 0 & 0 & 0 & z \\ 0 & 0 & 0 & z \\ 0 & 0 & 0 & z \\ z & z & z & 0 \end{pmatrix}$$

$$N_{-}^{(28)}(z) = \begin{pmatrix} 0 & 0 & z & 0 \\ 0 & z & 0 & 0 \\ z & 0 & 0 & 0 \\ 0 & 0 & 0 & z \end{pmatrix}$$

$$A_{+}(1) = \begin{pmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 2 & -1 \\ -1 & -1 & -1 & 2 \end{pmatrix}$$

$$A_{-}(1) = \begin{pmatrix} 2 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

 $h_{+} = 6, \quad h_{-} = 3, \quad \Omega = 18.$ Vertices per slice: 8.**Class 29** (constant ID 1494) $\mathbf{r} = (2, 2, 2, 2)$

$$N_{+}^{(29)}(z) = \begin{pmatrix} 0 & 0 & 0 & z \\ 0 & 0 & 0 & z \\ 0 & 0 & 0 & z \\ z & z & z & 0 \end{pmatrix}$$

$$N_{-}^{(29)}(z) = \begin{pmatrix} z & 0 & 0 & 0 \\ 0 & z & 0 & 0 \\ 0 & 0 & z & 0 \\ 0 & 0 & 0 & z \end{pmatrix}$$

$$A_{+}(1) = \begin{pmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 2 & -1 \\ -1 & -1 & -1 & 2 \end{pmatrix}$$

$$A_{-}(1) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

 $h_{+} = 6, \quad h_{-} = 3, \quad \Omega = 9.$ Vertices per slice: 8.

Classes 30–32

$$B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, \dots, 1 + z^{r_4}) - N_{\pm}^{(k)}(z).$$

Class 30 (constant ID 1834) $\mathbf{r} = (2, 2, 2, 2)$

$$N_+^{(30)}(z) = \begin{pmatrix} 0 & 0 & 0 & z \\ 0 & 0 & z & 0 \\ 0 & z & 0 & 0 \\ z & 0 & 0 & 0 \end{pmatrix}$$

$$N_-^{(30)}(z) = \begin{pmatrix} 0 & 0 & z & 0 \\ 0 & 0 & 0 & z \\ z & 0 & 0 & 0 \\ 0 & z & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & -1 & 0 \\ 0 & -1 & 2 & 0 \\ -1 & 0 & 0 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -1 & 0 \\ 0 & 2 & 0 & -1 \\ -1 & 0 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}$$

 $h_+ = 3, \quad h_- = 3, \quad \Omega = 12.$ Vertices per slice: 4.**Class 31** (constant ID 1836) $\mathbf{r} = (2, 2, 2, 2)$

$$N_+^{(31)}(z) = \begin{pmatrix} 0 & 0 & 0 & z \\ 0 & 0 & z & 0 \\ 0 & z & 0 & 0 \\ z & 0 & 0 & 0 \end{pmatrix}$$

$$N_-^{(31)}(z) = \begin{pmatrix} 0 & 0 & z & 0 \\ 0 & z & 0 & z \\ z & 0 & z & 0 \\ 0 & z & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & -1 & 0 \\ 0 & -1 & 2 & 0 \\ -1 & 0 & 0 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}$$

 $h_+ = 3, \quad h_- = 5, \quad \Omega = 16.$ Vertices per slice: 8.**Class 32** (constant ID 1844) $\mathbf{r} = (2, 2, 2, 6)$

$$N_+^{(32)}(z) = \begin{pmatrix} 0 & 0 & 0 & z \\ 0 & 0 & z & 0 \\ 0 & z & 0 & 0 \\ z^5 + z & 0 & z^3 & 0 \end{pmatrix}$$

$$N_-^{(32)}(z) = \begin{pmatrix} 0 & z & 0 & 0 \\ z & 0 & 0 & 0 \\ 0 & 0 & 0 & z \\ z^3 & 0 & z^5 + z & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & -1 & 0 \\ 0 & -1 & 2 & 0 \\ -2 & 0 & -1 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ -1 & 0 & -2 & 2 \end{pmatrix}$$

 $h_+ = 8, \quad h_- = 8, \quad \Omega = 16.$ Vertices per slice: 6.

Classes 33–35

$$B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, \dots, 1 + z^{r_4}) - N_{\pm}^{(k)}(z).$$

Class 33 (constant ID 1848)**r** = (2, 2, 2, 10)

$$N_+^{(33)}(z) = \begin{pmatrix} 0 & 0 & 0 & z \\ 0 & 0 & z & 0 \\ 0 & z & 0 & 0 \\ z^9 + z^5 + z & 0 & z^7 + z^3 & 0 \end{pmatrix}$$

$$N_-^{(33)}(z) = \begin{pmatrix} 0 & z & 0 & 0 \\ z & 0 & 0 & 0 \\ 0 & 0 & 0 & z \\ z^7 + z^3 & 0 & z^9 + z^5 + z & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & -1 & 0 \\ 0 & -1 & 2 & 0 \\ -3 & 0 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ -2 & 0 & -3 & 2 \end{pmatrix}$$

 $h_+ = 18, \quad h_- = 18, \quad \Omega = 36.$ Vertices per slice: 8.**Class 34** (constant ID 1852)**r** = (2, 2, 2, 2)

$$N_+^{(34)}(z) = \begin{pmatrix} 0 & 0 & 0 & z \\ 0 & 0 & z & 0 \\ 0 & z & 0 & z \\ z & 0 & z & 0 \end{pmatrix}$$

$$N_-^{(34)}(z) = \begin{pmatrix} 0 & 0 & z & 0 \\ 0 & 0 & 0 & z \\ z & 0 & z & 0 \\ 0 & z & 0 & z \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -1 & 0 \\ 0 & 2 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}$$

 $h_+ = 5, \quad h_- = 5, \quad \Omega = 20.$ Vertices per slice: 8.**Class 35** (constant ID 1853)**r** = (2, 2, 2, 2)

$$N_+^{(35)}(z) = \begin{pmatrix} 0 & 0 & 0 & z \\ 0 & 0 & z & 0 \\ 0 & z & 0 & z \\ z & 0 & z & 0 \end{pmatrix}$$

$$N_-^{(35)}(z) = \begin{pmatrix} z & 0 & 0 & 0 \\ 0 & z & 0 & 0 \\ 0 & 0 & z & 0 \\ 0 & 0 & 0 & z \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

 $h_+ = 5, \quad h_- = 3, \quad \Omega = 16.$ Vertices per slice: 8.

Classes 36–37

$$B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, \dots, 1 + z^{r_4}) - N_{\pm}^{(k)}(z).$$

Class 36 (constant ID 1902)**r** = (2, 2, 2, 3)

$$N_+^{(36)}(z) = \begin{pmatrix} 0 & 0 & 0 & z \\ 0 & 0 & z & 0 \\ 0 & z & z & 0 \\ z^2 + z & 0 & 0 & 0 \end{pmatrix}$$

$$N_-^{(36)}(z) = \begin{pmatrix} z & z & 0 & 0 \\ z & 0 & 0 & 0 \\ 0 & 0 & 0 & z \\ 0 & 0 & z^2 + z & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ -2 & 0 & 0 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & -2 & 2 \end{pmatrix}$$

 $h_+ = 5, \quad h_- = 5, \quad \Omega = 10. \quad \text{Vertices per slice: } 9.$
Class 37 (constant ID 2614)**r** = (2, 2, 6, 6)

$$N_+^{(37)}(z) = \begin{pmatrix} 0 & 0 & 0 & z \\ 0 & 0 & z & 0 \\ z^3 & z^5 + z & 0 & 0 \\ z^5 + z & z^3 & 0 & 0 \end{pmatrix}$$

$$N_-^{(37)}(z) = \begin{pmatrix} 0 & 0 & z & 0 \\ 0 & 0 & 0 & z \\ z^5 + z & z^3 & 0 & 0 \\ z^3 & z^5 + z & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & -1 & 0 \\ -1 & -2 & 2 & 0 \\ -2 & -1 & 0 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -1 & 0 \\ 0 & 2 & 0 & -1 \\ -2 & -1 & 2 & 0 \\ -1 & -2 & 0 & 2 \end{pmatrix}$$

 $h_+ = 12, \quad h_- = 12, \quad \Omega = 24. \quad \text{Vertices per slice: } 8.$

A Reproducibility and proof dependencies

This classification uses exact arithmetic and symbolic feasibility checks. The finite-type implication from simultaneous positivity, the coefficient bound, the enumeration scheme and the chain obstruction are proved above. The finite enumeration and the remaining feasibility and coverage statements are supported by executable exact-arithmetic checks. The computation trusts the C++ and Python implementations, SymPy rational arithmetic and Z3's UNSAT answers; it is not a proof-assistant formalization.

The checked runtime is Python 3.10.6, SymPy 1.14.0 and Z3 5.1.0. The source files are under `research/rank4/`. The mutation routines and installed Z3 package are reused from `research/rank3/`.

File	Content
<code>enumerate_constants.cpp</code>	Exhaustive ordered-vector enumeration.
<code>constant_candidates.json</code>	The 4,865 canonical constant pairs.
<code>constant_verification.json</code>	Integer positive vectors and parity exclusions.
<code>lift_feasibility.jsonl</code>	All lift decisions and the 37 witnesses.
<code>families.jsonl</code>	Linear equations and exact periodicity certificates.
<code>smt_queries.zip</code>	Complete exclusion and coverage formulas.
<code>verification.json</code>	Replayed answers, encodings, versions and hashes.
<code>slice_signatures.json</code>	Full canonical slice encodings.
<code>classification.json</code>	The catalogue, including both matrices at 1.

Run the commands in `REPRODUCE.md` from the repository root. The enumerator has no delay parameter. The lift solver preserves UNKNOWN answers and the verifier refuses a complete classification while any remain. The final replay contains 4,865 UNSAT answers: 4,407 have elementary parity or chain exclusions, 421 exclude the remaining impossible lifts, and 37 exclude lifts outside the displayed families.

Building this document. Run `python research/rank4/make_pdf_tables.py`, then build `rank4-classification.tex` with LuaLaTeX and `latexmk`, writing to `output/pdf/`. The generator evaluates every polynomial matrix at 1 and checks it against the separately enumerated constants.

References

- [1] Y. Mizuno, *Periodic Y-systems and Nahm sums: the rank 2 case*, SIGMA **21** (2025), 094, 17 pp. doi:10.3842/SIGMA.2025.094. arXiv:2301.13239v2.
- [2] Y. Mizuno, *Difference equations arising from cluster algebras*, arXiv:1912.05710v2 (2020).